Research Statement

Yu-You Liou

2026-08-07

[中文版由此去]

Air pollution, road deaths, and groundwater depletion share one feature. Whoever causes the damage does not pay for it. Environmental and resource economics calls this an externality, and policy is the attempted correction. My research measures how much of that correction arrives, and what else arrives with it. The answer is often not what the policy assumed. I use causal inference and large-scale administrative data. One line of work is on farms, the other on cities.

1. Land, Food, and the Environment

Agriculture is the largest user of land and water, a major source of emissions and nutrient runoff, and how millions of households live. A policy that cleans up farming at the farmer's expense will not be adopted. Must environmental quality and farm income trade off? Often not.

Solar power is treated as a rival for farmland. [1] and [2] show that panels on existing farm buildings displace nothing. Chicken farms that adopt them sell 5.8% more, from output rather than quality. The conflict is about siting, not solar.

Fertilizer and pesticide runoff resists regulation because it comes from tens of thousands of sources no inspector can reach. [3] finds that farms adopting food traceability are 8.2, 15.4, and 40 percentage points less likely to use chemical fertilizer, pesticide, and groundwater. Revenue rises too. A food safety system did what enforcement cannot.

Biodiversity disappears from private land because nobody pays for it. [4] finds that forest farms with greater tree diversity earn less but face far steadier revenue, and stability outweighs the loss. Diversity pays in stability, not income. Programs that compensate forgone income misread the asset.

2. Cities, Movement, and External Costs

Transport emissions keep rising, and transport kills directly. Cities answer with rail, fare subsidies, and traffic rules. None is cheap, and none has an obvious effect.

[5] uses the removal of the two-stage left turn rule for motorcycles in part of Tainan City. Accidents fell 21%, victims 19%, and vehicles involved 28%. Road safety usually means building something. Here a rule change cost nothing and did as well.

Fare subsidies are now sold as climate policy. [6] finds that the Taipei monthly pass raised ridership, passenger kilometers, and trip value by 4.3%, 4.6%, and 4.6%. Emissions fall only if 70% to 80% of the new trips come out of cars and motorcycles. Programs are judged on ridership. Ridership is the wrong number. [7] shows the demand is fragile as well, falling 1.43% per COVID-19 case.

Mass events send the bill to people who never attended. [8] finds accidents rose 14.9% and victims 17.4% during the Dajia Mazu Pilgrimage, and 9.8% even in townships off the route. [9] measures the air quality cost of Hakka tomb sweeping at the Lantern Festival. Neither cost enters the planning.

3. Other

[10] builds a strategic trade model with green investment. The Carbon Border Adjustment Mechanism pushes exporters toward cleaner technology, though its market structure effects vary by industry. [11] finds that consumers shifted food purchases online under pandemic risk. [12] finds that hotel revenue recovered through price rather than volume, with restored labor supply worth up to 17.8%.

Rural or urban, one question runs through all of it. When policy changes the price of environmental damage, who changes behavior, by how much, and who pays.

References

[1] Y.-Y. Liou, H.-H. Chang, and J. C. Begun. (2025). Rooftop Photovoltaics Adoption and Farm Revenue. *Agribusiness*. https://doi.org/10.1002/agr.70035

[2] T.-H. Lee, Y.-Y. Liou, and H.-H. Chang. (2025). Is Solar Panel Adoption a Win–Win Strategy for Chicken Farms? Evidence from Agriculture Census Data. *Agriculture* 15, (20), 2124. https://doi.org/10.3390/agriculture15202124

[3] H.-H. Chang, and Y.-Y. Liou. (2026). Food traceability systems and the probability of using chemical fertilizers and pesticides – empirical evidence using plot and farm household data from agriculture census. *Food Policy* 143, 103148. https://doi.org/10.1016/j.foodpol.2026.103148

[4] T.-H. Lee, Y.-Y. Liou, and H.-H. Chang. (2025). Forest diversity and the distribution of farm revenue - Empirical evidence from forest farms in Taiwan. *Forest Policy and Economics* 171, 103411. https://doi.org/10.1016/j.forpol.2024.103411

[5] Y.-Y. Liou, and H.-H. Chang. (2026). The causal effects of removing Hook-Turn regulation on road safety. *Transportation Research Part A: Policy and Practice* 205, 104860. https://doi.org/10.1016/j.tra.2026.104860

[6] Y.-Y. Liou, M.-K. Kim, H.-H. Chang, and R.-W. Liou. (2025). The causal effect of monthly pass programs on public transportation and carbon emissions. *Transportation Research Part D: Transport and Environment* 146, 104903. https://doi.org/10.1016/j.trd.2025.104903

[7] H.-H. Chang, B. Lee, F.-A. Yang, and Y.-Y. Liou. (2021). Does COVID-19 affect metro use in Taipei? *Journal of Transport Geography* 91, 102954. https://doi.org/10.1016/j.jtrangeo.2021.102954

[8] Y.-Y. Liou, and H.-H. Chang. (2026). Traffic accidents of religious tourism. *Annals of Tourism Research Empirical Insights* 7, (1), 100209. https://doi.org/10.1016/j.annale.2026.100209

[9] H.-H. Chang, Y.-Y. Liou, and C. D. Meyerhoefer. (2026). Air Pollution during Religious Holidays: Evidence from Tomb-Sweeping Day. *Journal of the Association of Environmental and Resource Economists*. Forthcoming. https://doi.org/10.1086/743452

[10] Y.-Y. Liou, H.-H. Chang, and J. C. Begun. (2025). The impact of the carbon border adjustment mechanism policy on international trade and market competition — A theoretical justification. *Singapore Economic Review*. https://doi.org/10.1142/S0217590825500328

[11] Y.-Y. Liou, H.-H. Chang, and D. R. Just. (2024). How do consumers respond to COVID-19? Application of Bayesian approach on credit card transaction data. *Quality & Quantity* 58, (6), 5737–5754. https://doi.org/10.1007/s11135-024-01915-9

[12] Y.-Y. Liou, M.-K. Kim, and H.-H. Chang. (2026). Hotel recovery strategies in the post-COVID-19 era: Evidence on revenue, employment, and labor market rigidity from Taiwan. *Journal of Tourism Management Research* 13, (1), 148–162. https://doi.org/10.18488/31.v13i1.4940